

INCHEON, Korea, Republic of

WMO: 471120

Lat: 37.4778N

Lon: 126.6250E

Elev: 227

StdP: 14.58

Time Zone: 9.00 (E09)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	13.3	16.9	-4.7	4.3	17.3	-1.4	5.2	19.9	25.1	21.5	21.6	25.1	9.4	0	0.402

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	9.8	88.8	77.6	86.3	76.5	84.1	75.1	80.6	84.4	79.1	83.5	77.6	82.0	6.6	270

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
79.8	155.8	82.4	77.9	146.0	81.4	76.2	138.1	80.4	44.6	84.2	42.9	83.5	41.3	82.2	86.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
19.5	16.5	14.3	9.3	92.0	4.3	2.7	6.2	94.0	3.7	95.5	1.3	97.1	-1.9	99.0		
			6.8	81.1	3.7	2.6	4.1	83.0	1.9	84.6	-0.2	86.0	-2.9	87.9		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	55.1	29.3	33.6	42.6	53.5	63.0	71.0	76.6	78.8	71.4	60.4	46.6	33.5
	DBStd	17.96	7.23	6.86	5.96	6.03	4.63	3.57	3.93	4.17	4.28	6.00	7.75	7.79
	HDD50	2041	641	459	239	31	0	0	0	0	0	7	152	512
	HDD65	4820	1106	879	694	349	95	3	0	0	5	161	552	976
	CDD50	3919	0	1	9	135	403	630	825	894	641	329	51	1
	CDD65	1221	0	0	0	3	32	183	360	429	196	18	0	0
	CDH74	7722	0	0	0	16	122	683	2390	3680	806	25	0	0
	CDH80	1814	0	0	0	3	9	86	563	1056	97	0	0	0
Wind	WSAvg	6.3	7.1	7.4	8.1	7.5	6.3	5.2	5.6	5.3	4.7	5.2	6.5	7.2
Precipitation	PrecAvg	46.2	0.7	0.9	1.5	3.0	3.4	4.4	11.6	10.1	5.5	2.0	2.0	0.9
	PrecMax	79.1	3.0	3.5	4.3	13.9	11.3	17.3	34.0	30.0	21.3	9.7	6.6	2.6
	PrecMin	25.6	0.1	0.0	0.1	0.3	0.3	0.2	3.1	1.1	0.0	0.1	0.2	0.0
	PrecStd	11.3	0.6	0.8	1.0	2.5	2.3	3.5	5.8	6.5	5.0	1.9	1.3	0.6
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	49.4	55.4	62.3	75.9	81.0	85.4	92.4	93.0	85.6	77.0	66.6	53.8
		MCWB	44.0	46.6	52.0	58.9	63.1	68.3	76.8	78.9	72.4	64.8	57.9	46.4
	2%	DB	44.5	49.3	58.0	69.7	77.4	82.4	87.7	90.0	82.8	73.7	62.9	49.9
		MCWB	39.6	42.3	48.9	56.1	62.1	68.2	77.2	78.5	71.0	63.5	55.7	44.2
	5%	DB	41.6	45.8	54.4	66.1	74.3	80.1	85.3	87.7	80.5	71.1	60.1	47.1
		MCWB	37.3	39.5	46.2	54.0	61.2	68.1	76.3	77.6	70.0	62.0	53.4	42.2
	10%	DB	39.2	43.0	51.3	63.1	71.5	77.9	83.0	85.5	78.5	68.8	57.7	44.6
		MCWB	35.0	37.9	44.2	52.5	60.2	67.8	75.6	76.8	68.8	60.8	52.0	39.9
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	45.1	50.5	55.6	63.0	68.0	74.6	81.9	83.9	77.2	69.1	60.9	49.0
		MCDB	48.1	53.4	59.1	69.7	73.7	77.8	85.0	86.9	80.9	73.1	64.6	51.9
	2%	WB	40.6	43.1	51.0	59.1	65.8	72.7	79.8	81.6	75.1	66.3	57.6	45.1
		MCDB	43.5	47.1	55.6	66.5	71.7	77.0	83.5	85.1	78.8	70.7	60.9	48.7
	5%	WB	38.1	40.7	47.7	56.6	64.2	71.4	78.3	80.0	73.5	64.5	55.2	42.7
		MCDB	41.1	44.9	52.3	63.0	70.2	76.0	82.4	84.3	77.1	68.7	58.9	46.6
	10%	WB	35.4	38.5	45.2	54.1	62.4	70.1	77.0	78.7	71.9	62.6	52.4	40.4
		MCDB	38.5	42.5	50.3	60.5	68.8	74.8	81.1	83.5	75.8	67.3	56.7	44.2
Mean Daily Temperature Range	5% DB	MDBR	11.0	11.8	12.3	13.1	13.5	11.9	9.2	9.8	12.2	13.2	11.9	11.2
		MCDBR	12.3	14.1	15.6	18.1	17.9	15.4	12.6	12.5	14.1	14.5	13.6	12.9
		MCWBR	9.6	9.5	9.3	8.7	7.9	6.2	5.7	5.3	6.4	7.6	9.7	10.3
	5% WB	MCDBR	11.5	12.9	13.6	15.2	14.3	11.5	10.0	10.2	10.7	12.5	11.7	12.2
MCWBR		10.0	9.9	9.7	9.2	8.4	6.2	5.6	5.5	6.2	8.4	10.3	11.1	
Clear-Sky Solar Irradiance	taub	0.433	0.485	0.562	0.607	0.621	0.694	0.662	0.577	0.508	0.500	0.481	0.433	
	taud	1.963	1.871	1.734	1.687	1.711	1.633	1.749	1.953	2.063	1.954	1.943	1.983	
	Ebn at Noon	222	227	222	222	222	206	212	228	236	220	205	211	
	Edh at Noon	43	54	68	75	75	81	71	57	49	49	44	40	
All-Sky Solar Radiation	RadAvg	799	1044	1377	1601	1797	1680	1323	1416	1336	1151	777	693	
	RadStd	50	83	133	105	171	187	206	199	140	80	75	45	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
(46)	Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(47)	Regional (44 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
										+167	+90

Nomenclature: See separate page